

(D)	The real root(s) of the equation $9^x - 10(3^x) + 9 = 0$ are	(S)	0, 2
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4.

(A)	If a, b are the roots of equation $x^2 + ax + b = 0$ ($a, b \in R$), then the number of ordered pairs (a, b) is equal to	(P)	1
(B)	If $P = \operatorname{cosec} \frac{\pi}{8} + \operatorname{cosec} \frac{2\pi}{8} + \operatorname{cosec} \frac{3\pi}{8} + \operatorname{cosec} \frac{13\pi}{8} + \operatorname{cosec} \frac{14\pi}{8} + \operatorname{cosec} \frac{15\pi}{8}$ and $Q = 8 \sin \frac{\pi}{18} \sin \frac{5\pi}{18} \sin \frac{7\pi}{18}$, then $P + Q$ is equal to	(Q)	2
(C)	Let $a_1, a_2, a_3, \dots$ be positive terms of a G.P. and $a_4, 1, 2, a_{10}$ are the consecutive terms of another G.P. If $\prod_{i=2}^{12} a_i = 4^{\frac{m}{n}}$ where m and n are coprime, then $(m + n)$ equals	(R)	3
(D)	For $x, y \in R$, if $x^2 - 2xy + 2y^2 - 6y + 9 = 0$, then the value of $5x - 4y$ is equal to	(S)	15

Answers

1.	$A \rightarrow Q; B \rightarrow P; C \rightarrow R; D \rightarrow S$
2.	$A \rightarrow Q; B \rightarrow S; C \rightarrow R; D \rightarrow P$
3.	$A \rightarrow Q; B \rightarrow R; C \rightarrow P; D \rightarrow S$
4.	$A \rightarrow Q; B \rightarrow P; C \rightarrow S; D \rightarrow R$

Exercise-5 : Subjective Type Problems

- Let $f(x) = ax^2 + bx + c$ where a, b, c are integers. If $\sin \frac{\pi}{7} \cdot \sin \frac{3\pi}{7} + \sin \frac{3\pi}{7} \cdot \sin \frac{5\pi}{7} + \sin \frac{5\pi}{7} \cdot \sin \frac{\pi}{7} = f\left(\cos \frac{\pi}{7}\right)$, then find the value of $f(2)$:
- Let a, b, c, d be distinct integers such that the equation $(x-a)(x-b)(x-c)(x-d) - 9 = 0$ has an integer root ' r ', then the value of $a + b + c + d - 4r$ is equal to :
- Consider the equation $(x^2 + x + 1)^2 - (m-3)(x^2 + x + 1) + m = 0$, where m is a real parameter. The number of positive integral values of m for which equation has two distinct real roots, is :
- The number of positive integral values of m , $m \leq 16$ for which the equation given in the above questions has 4 distinct real root is :
- If the equation $(m^2 - 12)x^4 - 8x^2 - 4 = 0$ has no real roots, then the largest value of m is $p\sqrt{q}$ where p, q are coprime natural numbers, then $p + q =$
- The least positive integral value of ' x ' satisfying $(e^x - 2) \left(\sin\left(x + \frac{\pi}{4}\right) \right) (x - \log_e 2) (\sin x - \cos x) < 0$ is :
- The integral values of x for which $x^2 + 17x + 71$ is perfect square of a rational number are a and b , then $|a - b| =$
- Let $P(x) = x^6 - x^5 - x^3 - x^2 - x$ and $\alpha, \beta, \gamma, \delta$ are the roots of the equation $x^4 - x^3 - x^2 - 1 = 0$, then $P(\alpha) + P(\beta) + P(\gamma) + P(\delta) =$
- The number of real values of ' a ' for which the largest value of the function $f(x) = x^2 + ax + 2$ in the interval $[-2, 4]$ is 6 will be :
- The number of all values of n , (where n is a whole number) for which the equation $\frac{x-8}{n-10} = \frac{n}{x}$ has no solution.
- The number of negative integral values of m for which the expression $x^2 + 2(m-1)x + m + 5$ is positive $\forall x > 1$ is :
- If the expression $ax^4 + bx^3 - x^2 + 2x + 3$ has the remainder $4x + 3$ when divided by $x^2 + x - 2$, then $a + 4b = \dots$
- Find the smallest value of k for which both the roots of equation $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values atleast 4.
- If $x^2 - 3x + 2$ is a factor of $x^4 - px^2 + q = 0$, then $p + q =$
- The sum of all real values of k for which the expression $x^2 + 2xy + ky^2 + 2x + k = 0$ can be resolved into linear factors is :
- The curve $y = (a+1)x^2 + 2$ meets the curve $y = ax + 3$, $a \neq -1$ in exactly one point, then $a^2 =$

17. Find the number of integral values of 'a' for which the range of function $f(x) = \frac{x^2 - ax + 1}{x^2 - 3x + 2}$ is $(-\infty, \infty)$.
18. When x^{100} is divided by $x^2 - 3x + 2$, the remainder is $(2^{k+1} - 1)x - 2(2^k - 1)$, then $k =$
19. Let $P(x)$ be a polynomial equation of least possible degree, with rational coefficients, having $\sqrt[3]{7} + \sqrt[3]{49}$ as one of its roots. Then the product of all the roots of $P(x) = 0$ is :
20. The range of values k for which the equation $2\cos^4 x - \sin^4 x + k = 0$ has atleast one solution is $[\lambda, \mu]$. Find the value of $(9\mu + \delta)$.
21. Let $P(x)$ be a polynomial with real coefficient and $P(x) - P'(x) = x^2 + 2x + 1$. Find $P(1)$.
22. Find the smallest positive integral value of a for which the greater root, of the equation $x^2 - (a^2 + a + 1)x + a(a^2 + 1) = 0$ lies between the roots of the equation $x^2 - a^2x - 2(a^2 - 2) = 0$
23. If the equation $x^4 + kx^2 + k = 0$ has exactly two distinct real roots, then the smallest integral value of $|k|$ is :
24. Let a, b, c, d be the roots of $x^4 - x^3 - x^2 - 1 = 0$. Also consider $P(x) = x^6 - x^5 - x^3 - x^2 - x$, then the value of $P(a) + P(b) + P(c) + P(d)$ is equal to :
25. The number of integral values of $a, a \in [-5, 5]$ for which the equation $x^2 + 2(a-1)x + a + 5 = 0$ has one root smaller than 1 and the other root greater than 3 is :
26. The number of non-negative integral values of $n, n \leq 10$ so that a root of the equation $n^2 \sin^2 x - 2 \sin x - (2n + 1) = 0$ lies in interval $\left[0, \frac{\pi}{2}\right]$ is :
27. Let $f(x) = ax^2 + bx + c$, where a, b, c are integers and $a > 1$. If $f(x)$ takes the value p , a prime for two distinct integer values of x , then the number of integer values of x for which $f(x)$ takes the value $2p$ is :
28. If x and y are real numbers connected by the equation $9x^2 + 2xy + y^2 - 92x - 20y + 244 = 0$, then the sum of maximum value of x and the minimum value of y is :
29. Consider two numbers a, b , sum of which is 3 and the sum of their cubes is 7. Then sum of all possible distinct values of a is :
30. If $y^2(y^2 - 6) + x^2 - 8x + 24 = 0$ and the minimum value of $x^2 + y^4$ is m and maximum value is M ; then find the value of $M - 2m$.
31. Consider the equation $x^3 - ax^2 + bx - c = 0$, where a, b, c are rational number, $a \neq 1$. It is given that x_1, x_2 and $x_1 x_2$ are the real roots of the equation. If $(b + c) = 2(a + 1)$, then $x_1 x_2 \left(\frac{a+1}{b+c} \right) =$
32. Let α satisfy the equation $x^3 + 3x^2 + 4x + 5 = 0$ and β satisfy the equation $x^3 - 3x^2 + 4x - 5 = 0, \alpha, \beta \in \mathbb{R}$, then $\alpha + \beta =$

33. Let x, y and z are positive reals and $x^2 + xy + y^2 = 2; y^2 + yz + z^2 = 1$ and $z^2 + zx + x^2 = 3$.
If the value of $xy + yz + zx$ can be expressed as $\sqrt{\frac{p}{q}}$ where p and q are relatively prime positive integer find the value of $p - q$:
34. The number of ordered pairs (a, b) , where a, b are integers satisfying the inequality $\min(x^2 + (a - b)x + (1 - a - b)) > \max(-x^2 + (a + b)x - (1 + a + b)) \forall x \in R$, is :
35. The real value of x satisfying $\sqrt[3]{20x + \sqrt[3]{20x + 13}} = 13$ can be expressed as $\frac{a}{b}$ where a and b are relatively prime positive integers. Find the value of b ?
36. If the range of the values of a for which the roots of the equation $x^2 - 2x - a^2 + 1 = 0$ lie between the roots of the equation $x^2 - 2(a + 1)x + a(a - 1) = 0$ is (p, q) , then find the value of $\left(q - \frac{1}{p}\right)$.
37. Find the number of positive integers satisfying the inequality $x^2 - 10x + 16 < 0$.
38. If $\sin \theta$ and $\cos \theta$ are the roots of the quadratic equation $ax^2 + bx + c = 0$ ($ac \neq 0$). Then find the value of $\frac{b^2 - a^2}{ac}$.
39. Let the inequality $\sin^2 x + a \cos x + a^2 \geq 1 + \cos x$ is satisfied $\forall x \in R$, for $a \in (-\infty, k_1] \cup [k_2, \infty)$, then $|k_1| + |k_2| =$
40. α and β are roots of the equation $2x^2 - 35x + 2 = 0$. Find the value of $\sqrt{(2\alpha - 35)^3 (2\beta - 35)^3}$
41. The sum of all integral values of ' a ' for which the equation $2x^2 - (1 + 2a)x + 1 + a = 0$ has a integral root.
42. Let $f(x)$ be a polynomial of degree 8 such that $F(r) = \frac{1}{r}, r = 1, 2, 3, \dots, 8, 9$, then $\frac{1}{F(10)} =$
43. Let α, β are two real roots of equation $x^2 + px + q = 0, p, q \in R, q \neq 0$. If the quadratic equation $g(x) = 0$ has two roots $\alpha + \frac{1}{\alpha}, \beta + \frac{1}{\beta}$ such that sum of its roots is equal to product of roots, then then number of integral values q can attain is :
44. If $\cos A, \cos B$ and $\cos C$ are the roots of cubic $x^3 + ax^2 + bx + c = 0$, where A, B, C are the angles of a triangle then find the value of $a^2 - 2b - 2c$.
45. Find the number of positive integral values of k for which $kx^2 + (k - 3)x + 1 < 0$ for atleast one positive x .

Answers

1. 9	2. 0	3. 1	4. 7	5. 5	6. 3	7. 3	8. 6	9. 0	10. 6
11. 0	12. 9	13. 2	14. 9	15. 2	16. 4	17. 0	18. 99	19. 56	20. 7
21. 2	22. 3	23. 1	24. 6	25. 4	26. 8	27. 0	28. 7	29. 3	30. 4
31. 1	32. 0	33. 5	34. 9	35. 5	36. 5	37. 5	38. 2	39. 3	40. 8
41. 1	42. 5	43. 3	44. 1	45. 0					

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